Definition: The sample space, S, of a stochastic experiment is the set in which every possible outcome of the experiment is represented by exactly one element, $\{s_1, s_2, \ldots\}$.

Definition: An event is a subset E of the sample space S, representing the collection of "favourable" outcomes corresponding to that event. If the stochastic experiment generates an outcome $O \in E$ we say that the event E occurs. An event that is represented by a set of only one outcome is called an elementary event. Events consisting of more than one outcome are called compound events.

Definition: Consider a stochastic experiment with sample space S. The experiment is repeated n times, and on each trial an event of interest $E \subseteq S$ occurs or doesn't occur. The relative frequency of the event, denoted $rf_n(E)$ is the number of times the event E occurs divided by the number of repetitions (or trials), n.

To define relative frequency mathematically, let $O_i = 1$ if the outcome of the ith trial is in E (that is, the event E occurs on the ith trial), and $O_i = 0$ if E does not happen on the ith trial. Then

$$rf_n(E) = \frac{\sum_{i=1}^{n} O_i}{n}$$

Definition: The probability of an event E, written $P(E)$, is the relative frequency of E in the limit of infinite n:

$$P(E) = \lim_{n \to \infty} rf_n(E)$$

$P(S) = 1$

$0 \leq P(E) \leq 1$

If events $E_1, E_2, \ldots E_K$ are mutually disjoint, then

$$P(E_1 \cup E_2 \cup \ldots \cup E_K) = P(E_1) + P(E_2) + \ldots + P(E_K)$$

$P(\emptyset) = 0$

$P(E^c) = 1 - P(E)$

Theorem. Probability Inclusion-Exclusion Principle (IEP): Let E_1 and E_2 be events, not necessarily disjoint. Then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

Theorem. Probability of Equally Likely Outcomes: Define the sample space $S = \{O_1, \ldots O_k\}$, and the corresponding elementary events $S_i = \{O_i\}$. If the k outcomes are equally likely outcomes, then $P(S_i) = \frac{1}{k}$ for all i.

Corollary: For any event $E \subseteq S$ associated with a sample space S of k equally likely outcomes, we have

$$P(E) = \frac{|E|}{k}$$

Definition: Suppose that E and F are events in a sample space, S, with $P(F) > 0$. The conditional probability of E given F, denoted $P(E|F)$, is defined as

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

Theorem. Multiplication Theorem: For any events E and F with $P(F) > 0$, we have

$$P(E|F) + P(E^c|F) = 1$$

Corollary: Let $E_1, E_2, \ldots, E_m$ be events such that $P(E_{m-1} \cap \ldots \cap E_2 \cap E_1) > 0$, then

$$P(E_m \cap \ldots \cap E_1) = P(E_m | E_{m-1} \cap \ldots \cap E_1) \times P(E_{m-1} | E_{m-2} \cap \ldots \cap E_1) \times \ldots \times P(E_2 | E_1) \times P(E_1)$$

Definition: The two events E and F are to be independant if and only if

$$P(E \cap F) = P(E)P(F)$$

Definition: The k events $E_1, \ldots, E_k$ are said to be pairwise independant if, each pair of events E_i and E_j where $i \neq j$ are independant. The k events $E_1, \ldots, E_k$ are said to be mutually (jointly) independant (or just say independant) if, all subsets of events are independant including

$$P(E_1 \cap \ldots \cap E_k) = P(E_1)P(E_2)\ldots P(E_k)$$

Theorem. Law of Total Probability (LOTP): Let the events $E_1, E_2, \ldots$ form a partition of the sample space S, and let A be an event $A \subseteq S$. Then

$$P(A) = \sum_i P(A|E_i)P(E_i)$$

Theorem. Bayes' Theorem: Let E and F be events in the sample space S with $P(F) > 0$. Then

$$P(E|F) = \frac{P(F|E)P(E)}{P(F)}$$